



SafeCross

AI-Based Smart Crosswalk System · HIT Project

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Real-Time Hazard Detection • Behavioral Risk Classification • Live Alerts • Safety Analytics

Solution

An AI-powered web platform that detects dangerous and unpredictable pedestrian behavior in real time and alerts a control center instantly.

Problem

Crosswalks are monitored passively: CCTV is used only after accidents, and unsignalized crossings rely on the driver's alertness alone.

Our methodology combines Python & YOLOv8 computer vision that tracks every pedestrian and classifies behavior into 17 risk cases, a Node.js & Express REST API with Socket.io real-time events, a MongoDB Atlas cloud database, and a responsive React & Vite web app with four role-based dashboards.



React · Vite · Tailwind CSS · Socket.io



Node.js · Express · MongoDB Atlas · Python/YOLOv8 · Cloudinary

- ❖ Secure login with JWT and role-based access
- ❖ Admin: users and hardware management (junctions, cameras, LEDs)
- ❖ Automatic address-to-coordinates (geocoding)
- ❖ Live synchronization across all screens (Socket.io)
- ❖ Dispatcher: live alert table and interactive map
- ❖ Manager: server-side analytics (Top 5, school zones, weekly trend) and PDF report export
- ❖ Technician: faulty-equipment list and repair status
- ❖ AI: YOLOv8 detection, tracking and 17 risk cases

The goal of the project is to transform passive crosswalks into active safety zones: detect dangerous and unpredictable pedestrian behavior – especially of children and distracted pedestrians – in real time, alert operators within seconds, and give municipalities reliable data to improve high-risk intersections.

SafeCross shows that an end-to-end real-time system combining computer vision, an event-driven backend and role-based dashboards can shorten response times, reduce false alarms through behavioral classification and human verification, and support data-driven road-safety decisions. Next steps: physical LED warning-sign control, live camera streams and learning from operator feedback.